What is the point of all this?

. ; : !

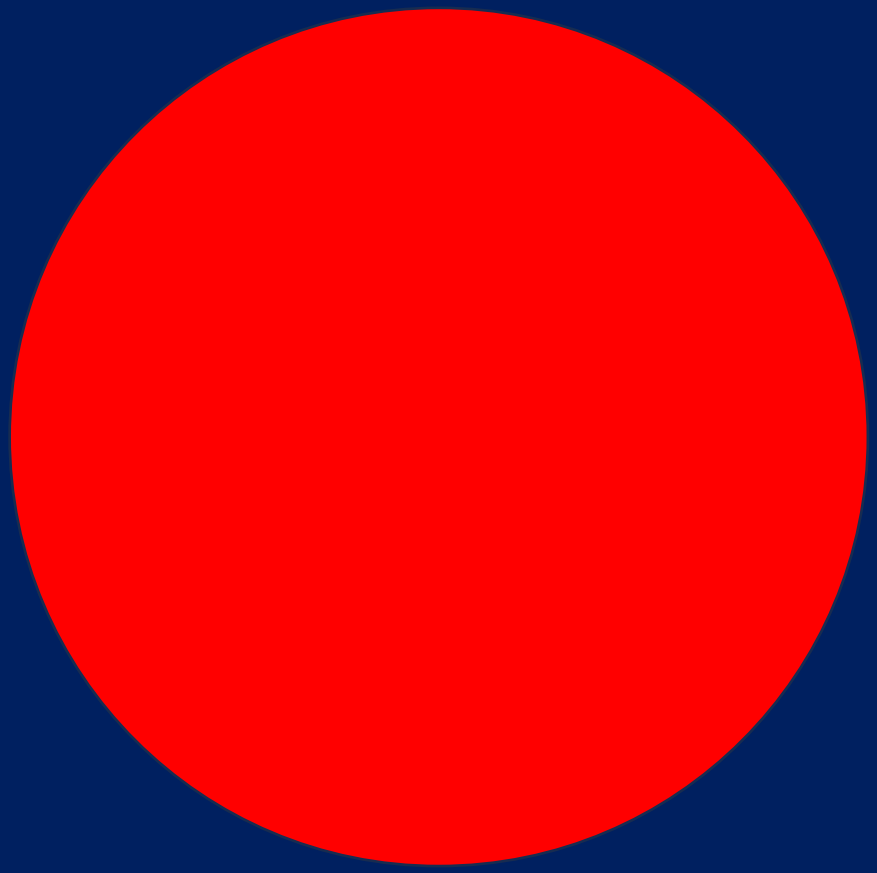
Scaling is foundational to making our science useful

Ankur Desai, UW-Madison

Remote Sensing and Fluxes Upscaling for Real-world Impact

Mar 2026

Point 1. There is no point

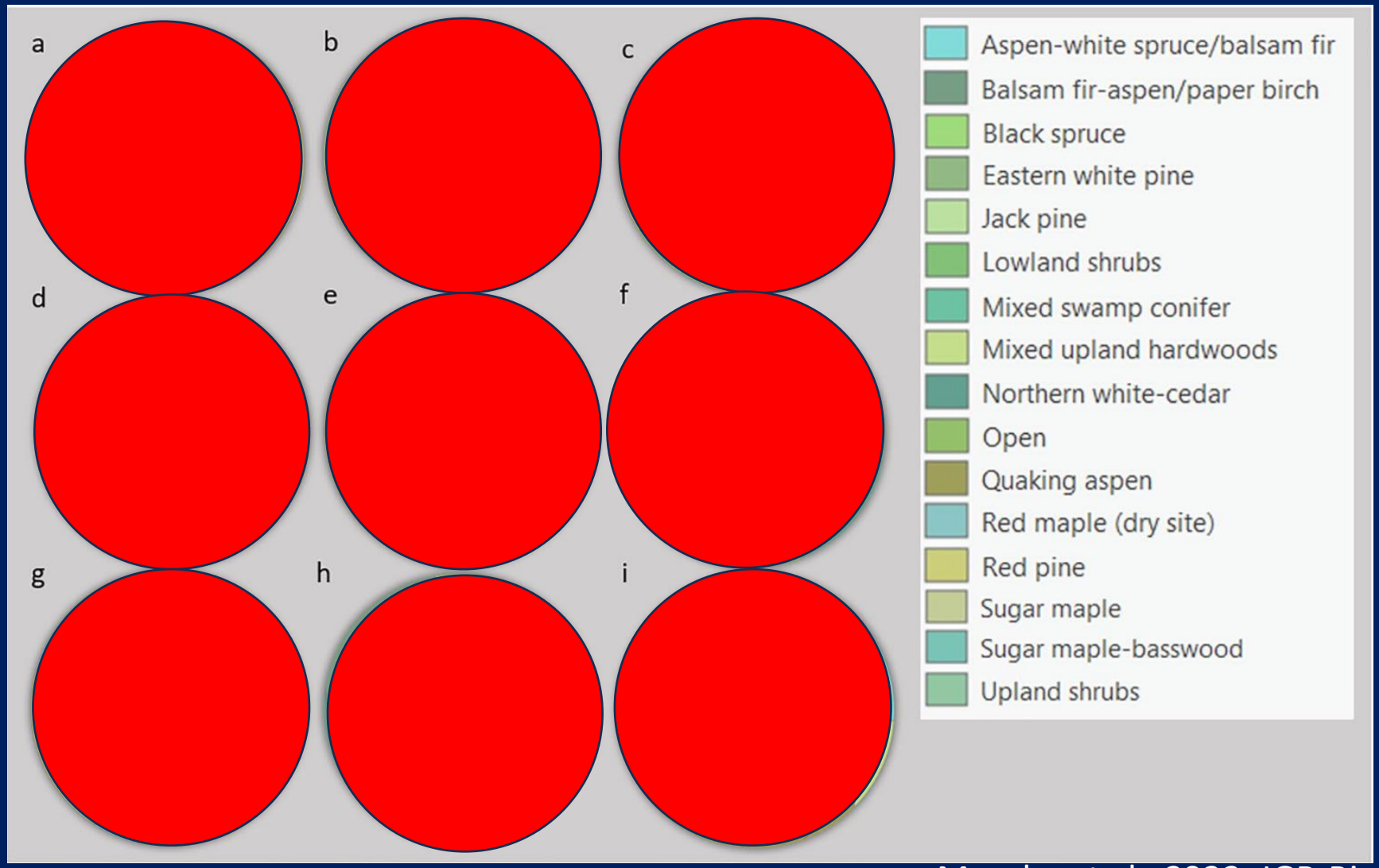


The points are areas!

And not a lot of area

- ~1,000 flux towers each sample a footprint of ~1 km^2 = 1,000 km^2
- Global terrestrial land area is ~150,000,000 km^2
- Flux towers sample 0.0007% of globe
- Similar story for chambers
- Similar story even for satellites

The points are not the same

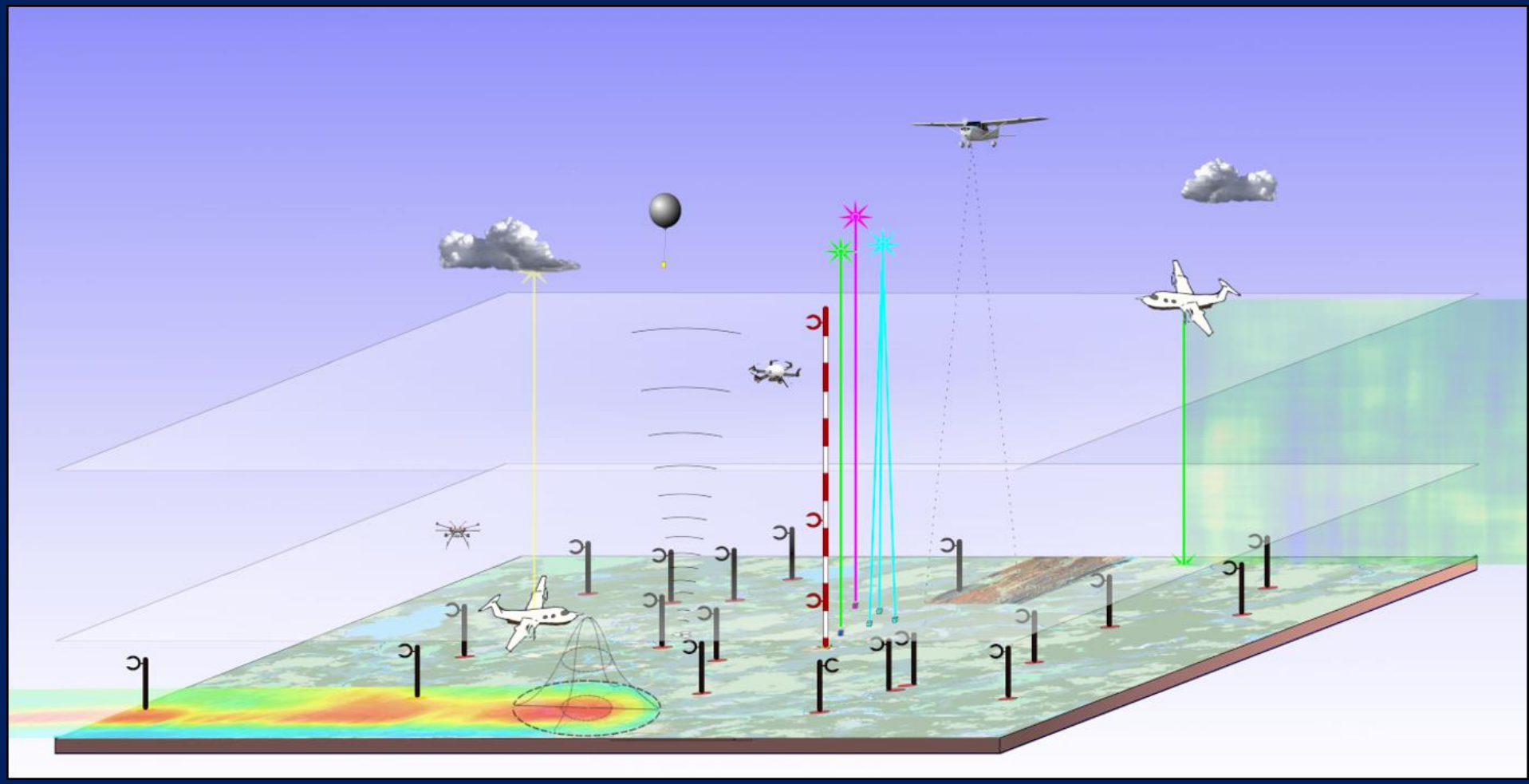


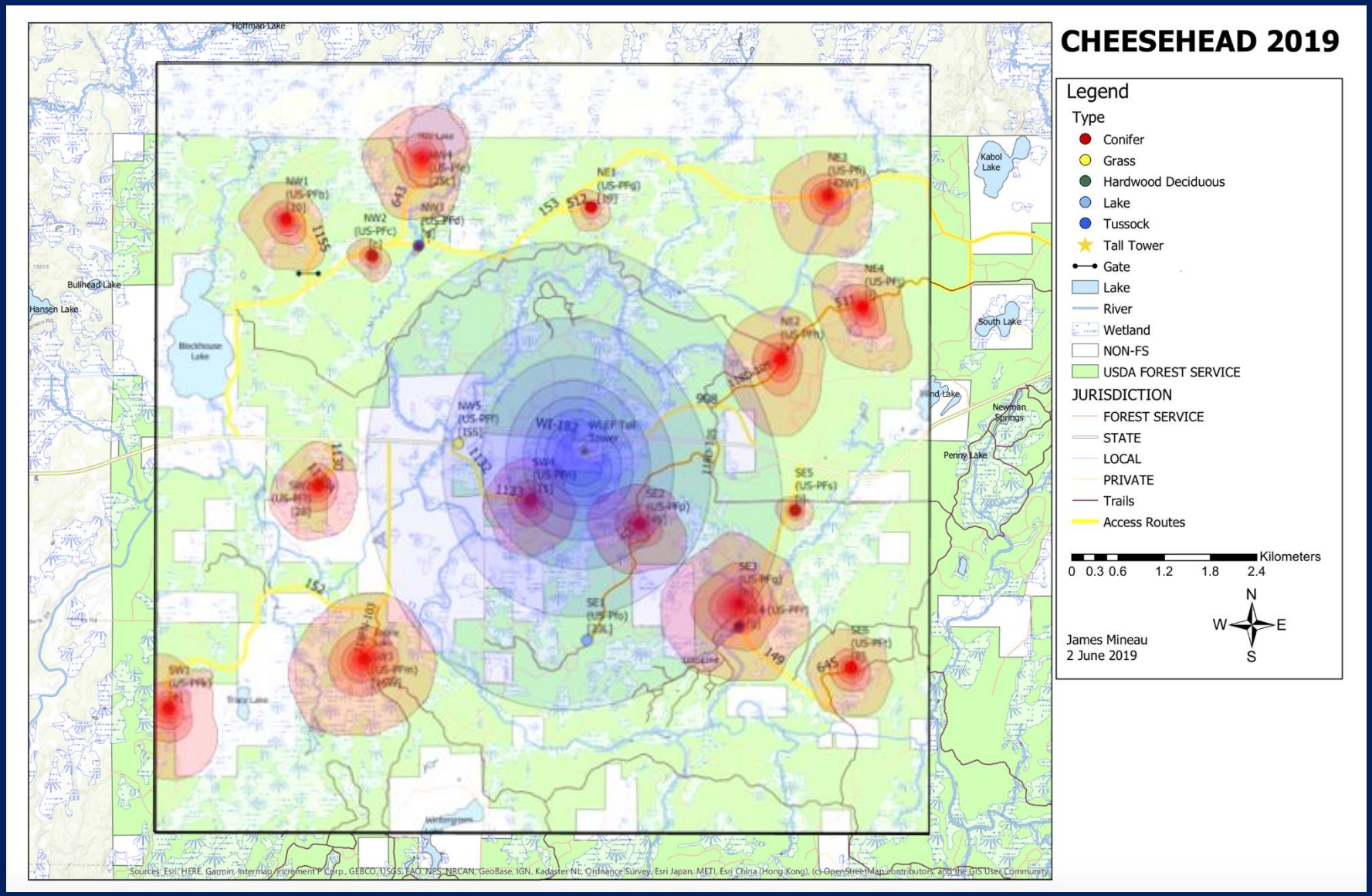
Murphy et al., 2022, JGR-Biogeo

Point 2. We are not pointillists

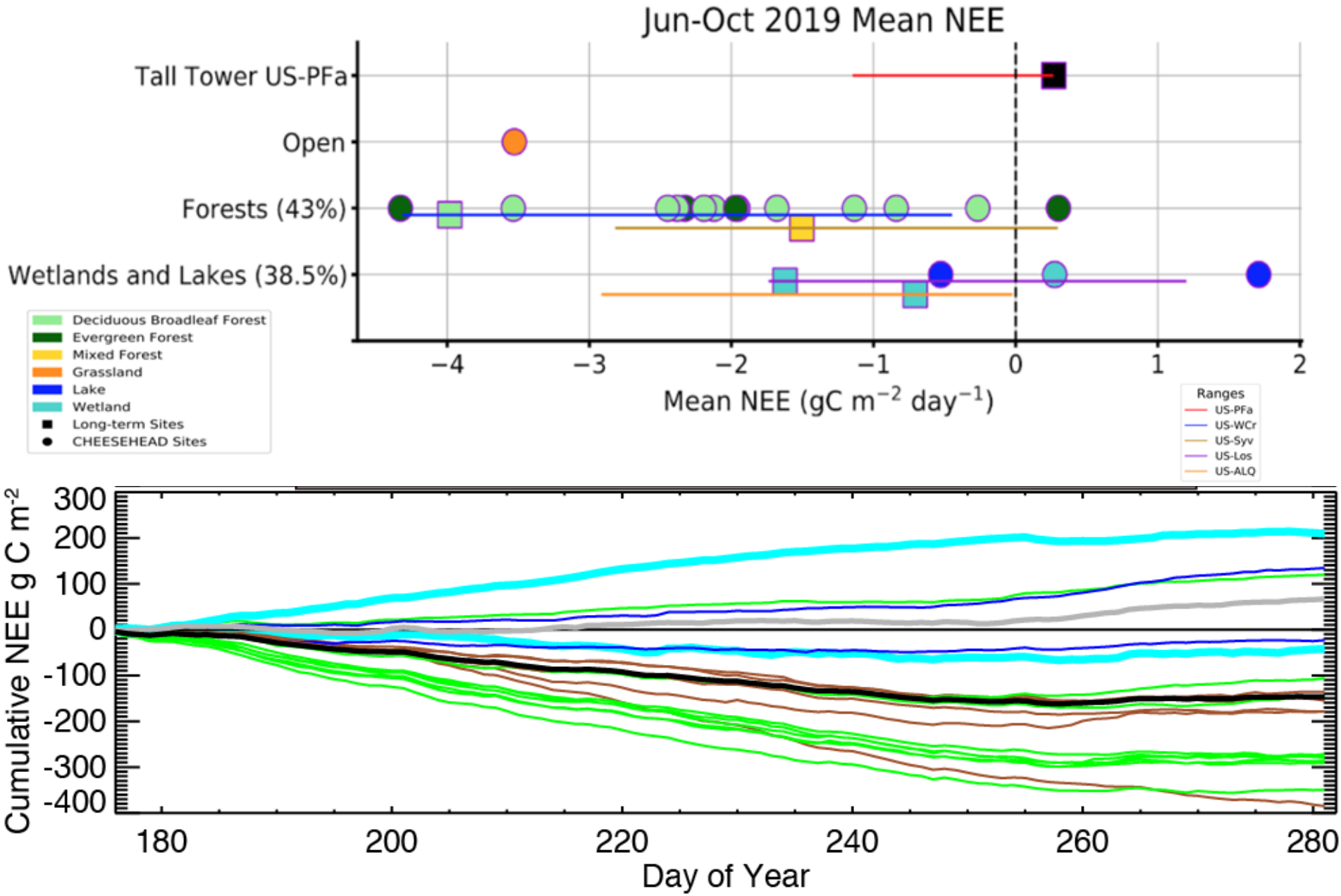
Seurat

How many points are enough?





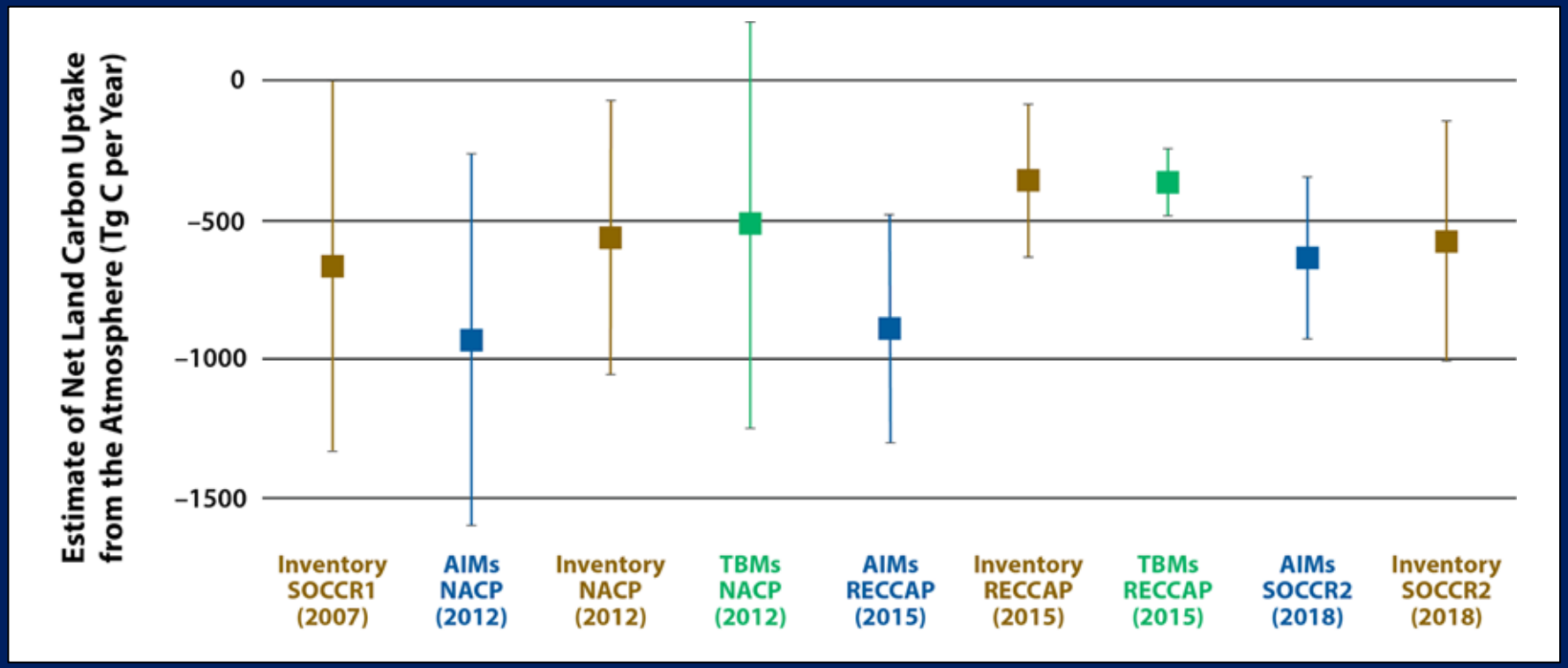
Butterworth et al., 2021



Desai et al., 2022, JGR-Biogeo

Point 3. We can never acquire enough points to win

Top-down and bottom-up convergence remain far apart



Anna Michalk, Poulter et al., 2025, GBC

This complicates any regional-global greenhouse gas policy

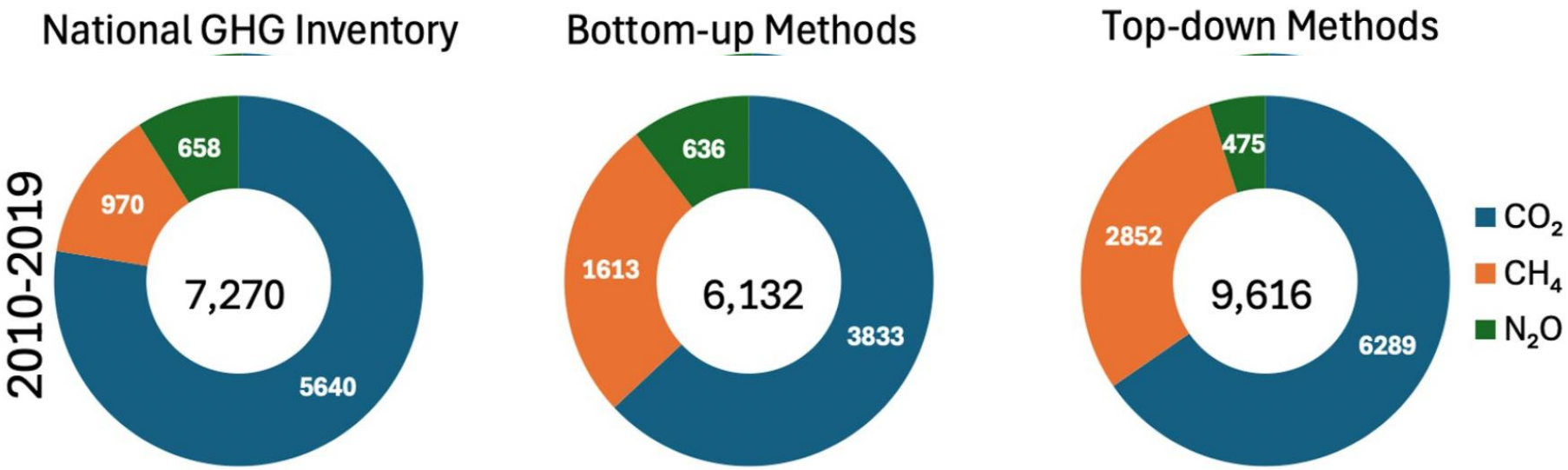
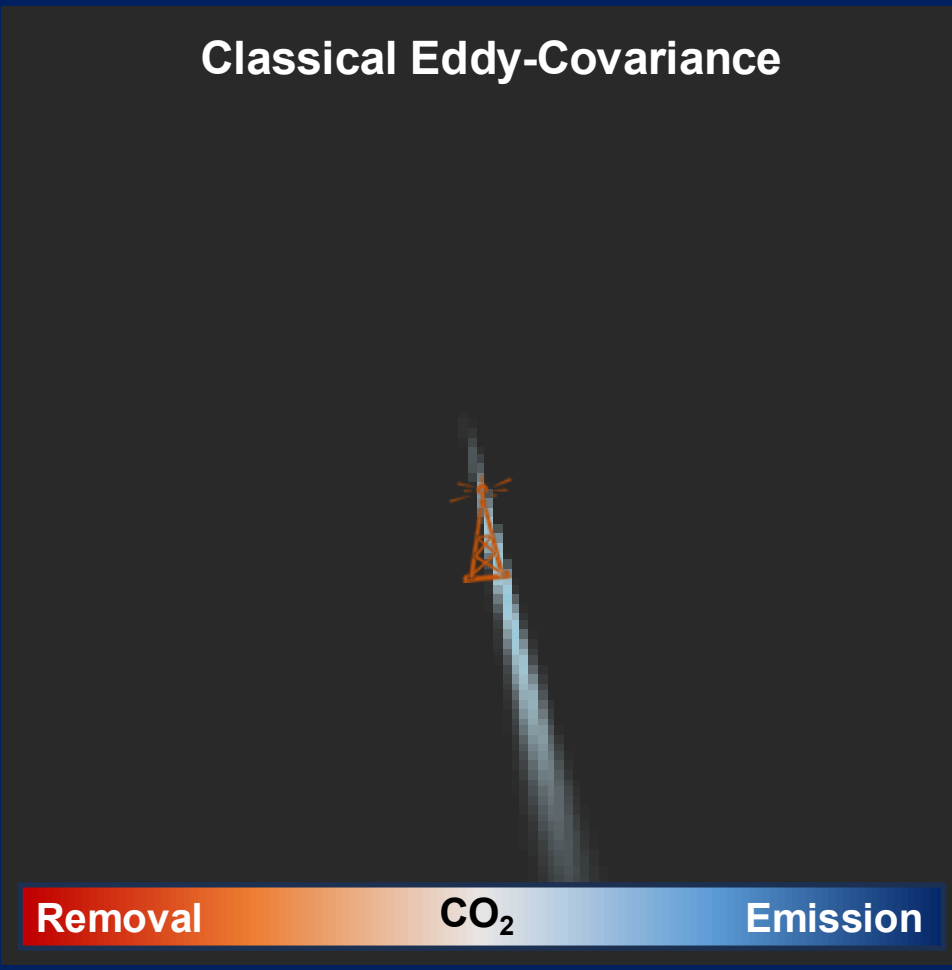
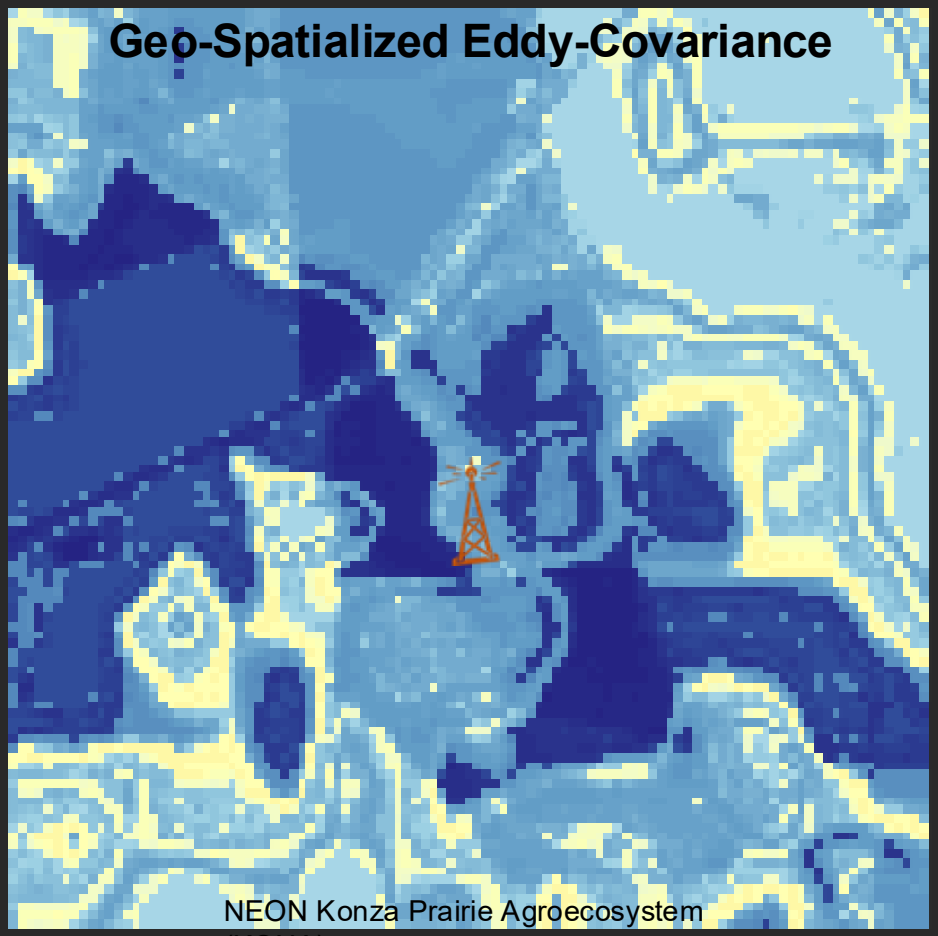


Figure 14. Distribution of anthropogenic, managed-lands and natural net greenhouse gas emissions ($TgCO_2$-eq yr^{-1}) for the entire North American region for 2000–2009 (upper panel) and 2010–2019 (lower panel). The columns (from left to right) contrast the national greenhouse gas inventory methods with the bottom-up and top-down methods.

Anna Michalk, Poulter et al., 2025, GBC

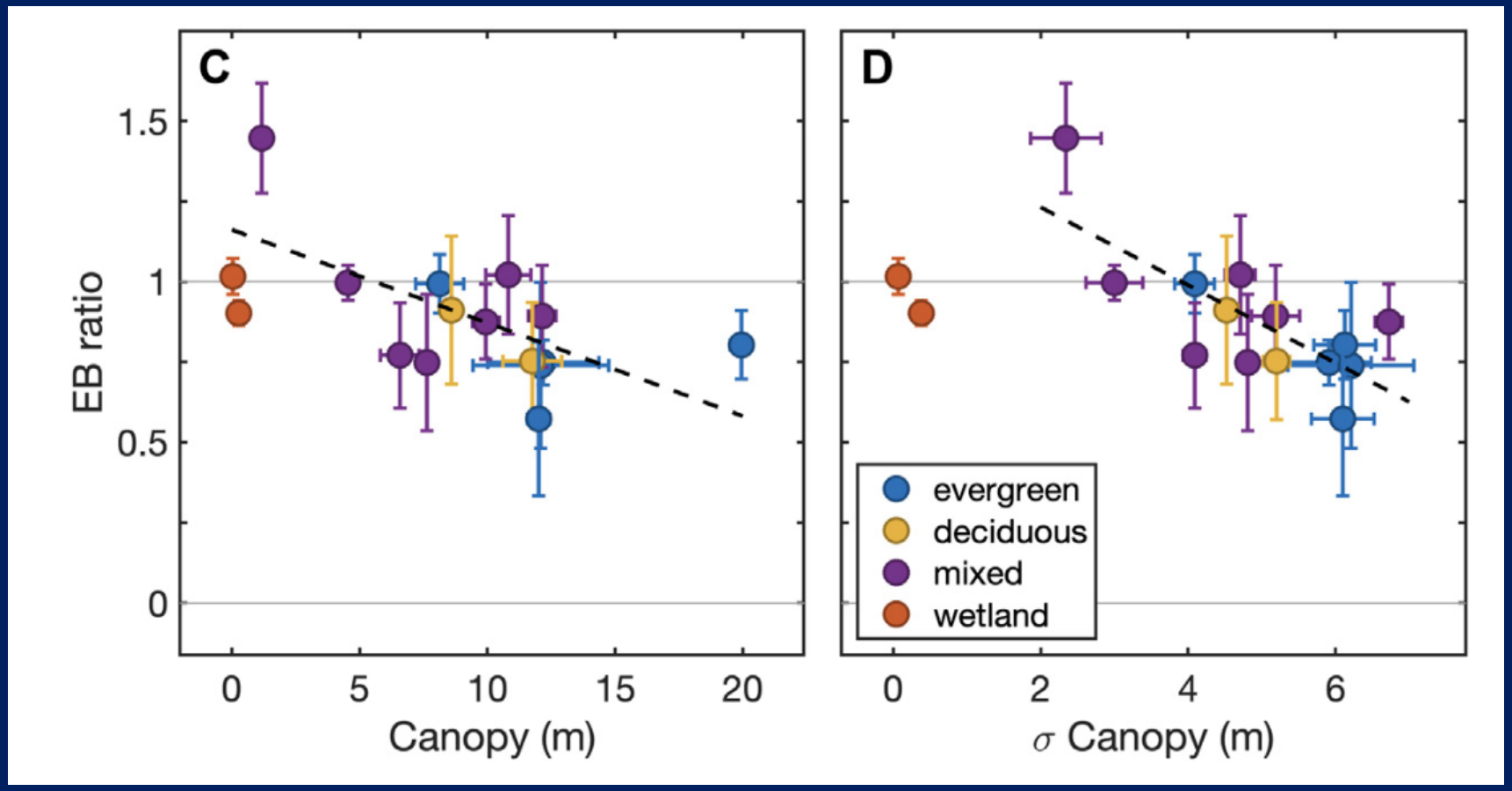
Point 4. We struggle with getting to the point





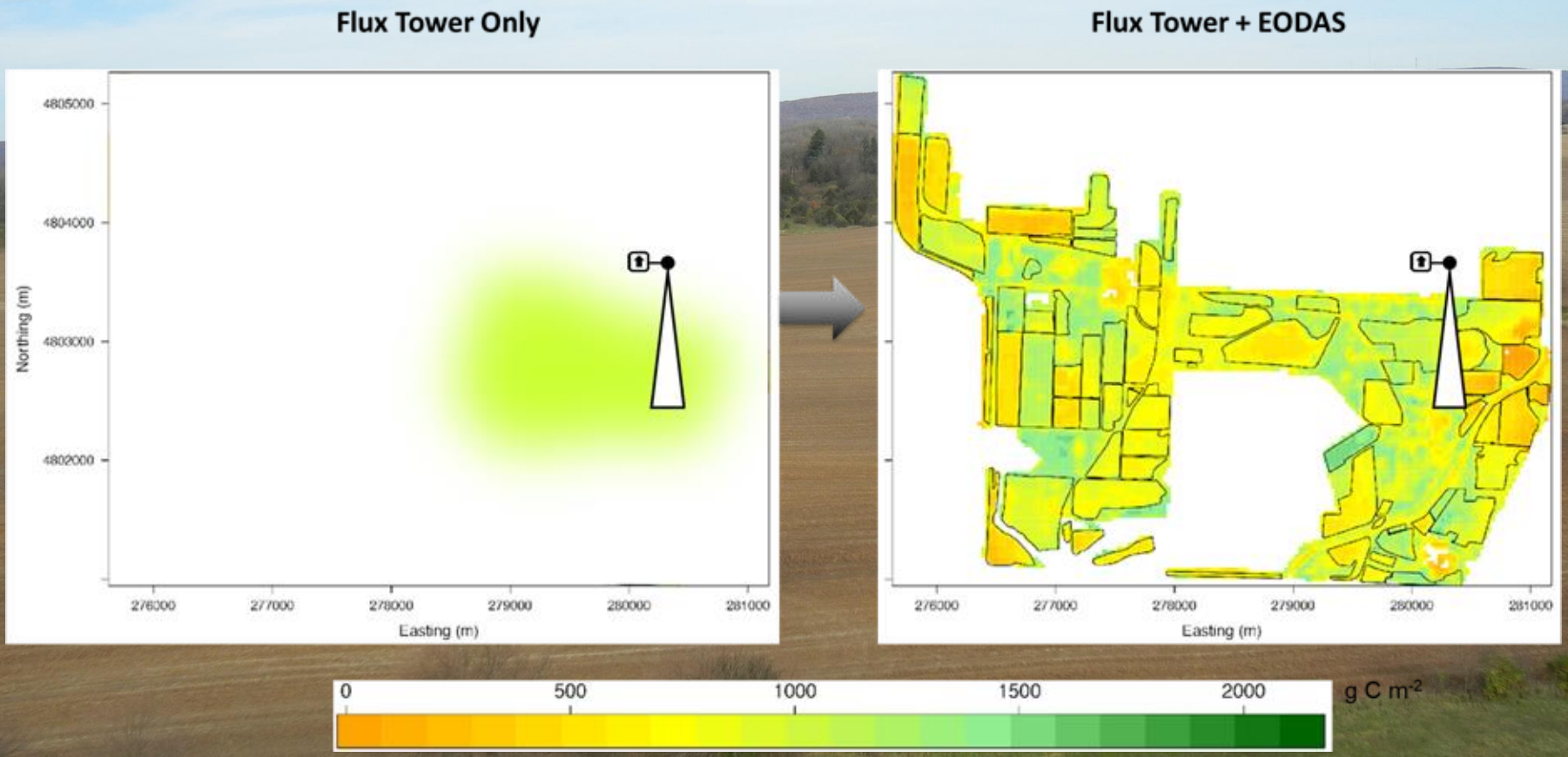
S. Metzger, Atmofacts + Benchflux (Google)

The points systematically lie to us



Butterworth et al., 2024

The “point” is not the point!



S Metzger + Wiesner et al., 2022, JGR-Biogeo

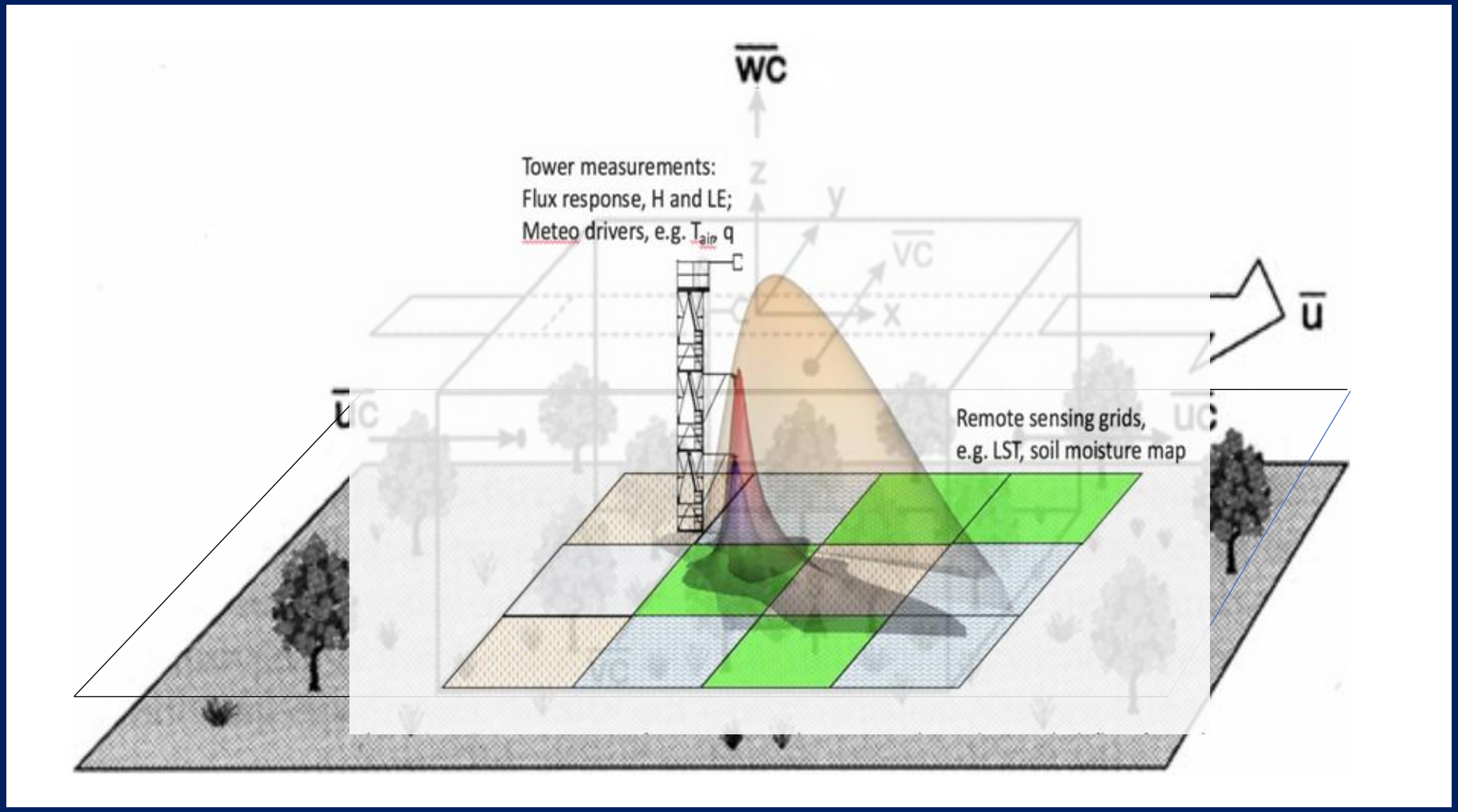
How do we move forward?

- Smarter scaling
  - Being "scale-aware"
  - Using modern ML techniques to account for it
- Better data integration
  - Across bottom-up and top-down observations (including top-down)
  - Field, management, intervention scale integration
- Uncertainty Everywhere All The Time
  - Mechanistic correction to energy imbalance
  - Propagation of uncertainty

Dumbed Down Dictionary for Desai

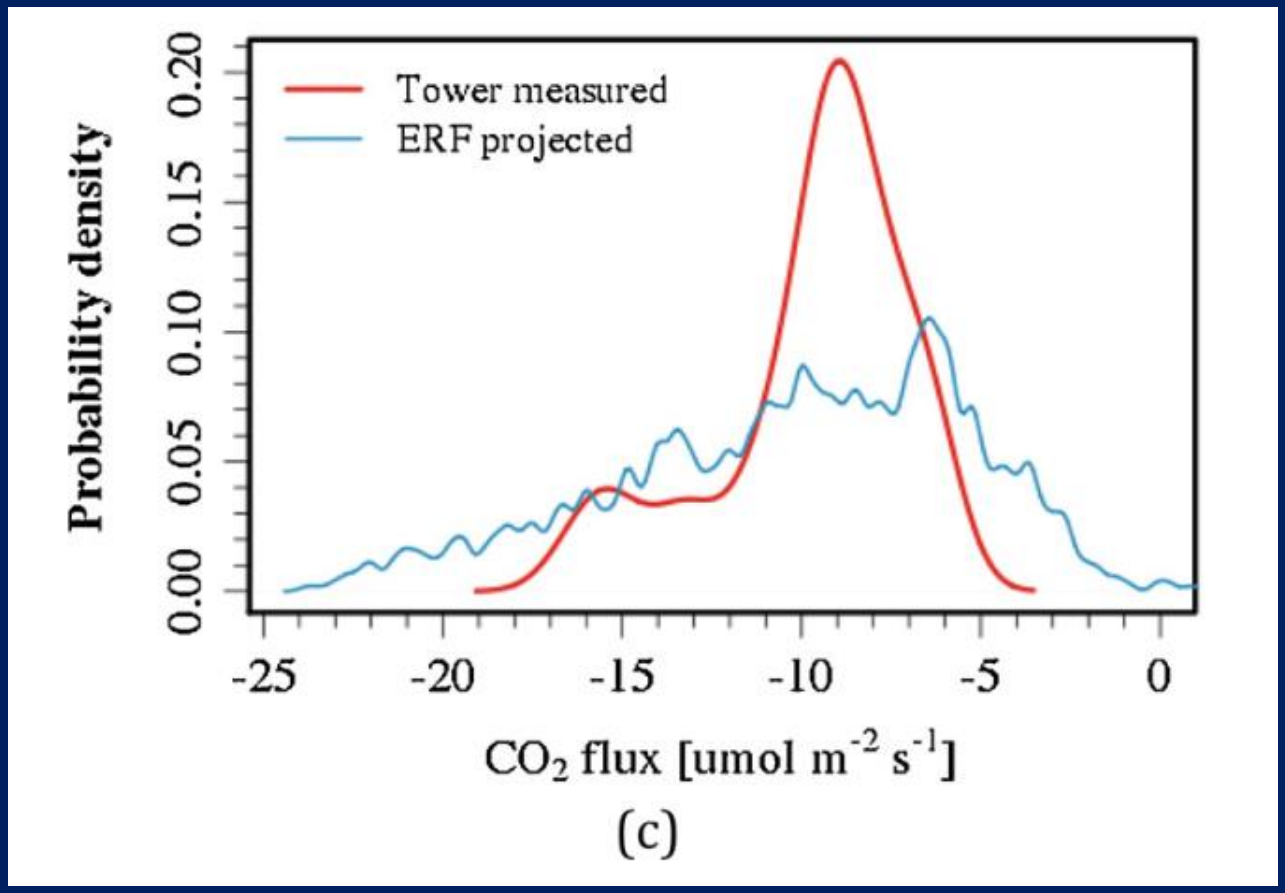
- Flux mapping
- Knowledge Guided Machine Learning (KGML)
- Foundation model
- Scale-aware
- Top down (downscale) and bottom up (upscale)
- Scale invariance and scale dependence
- Agentic AI
- AI-ready data
- Model context protocol
- Benchmark
- Vibe coding
- Explainable AI (xAI), Interpretability
- Hyper-

We can be scale-aware



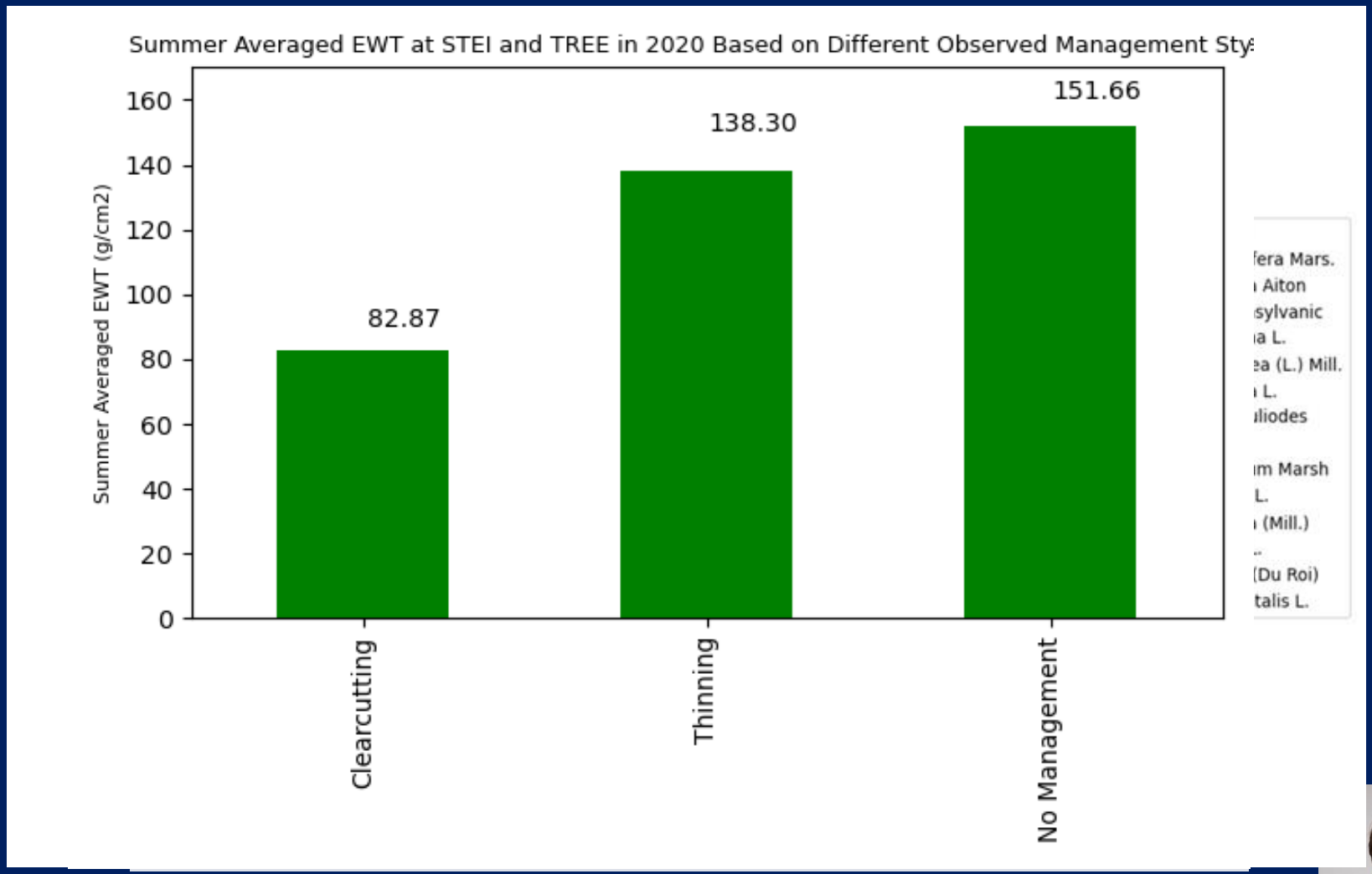
Finnigan et al (2003)

And use scale-awareness to build better estimates of spatial variation and uncertainty



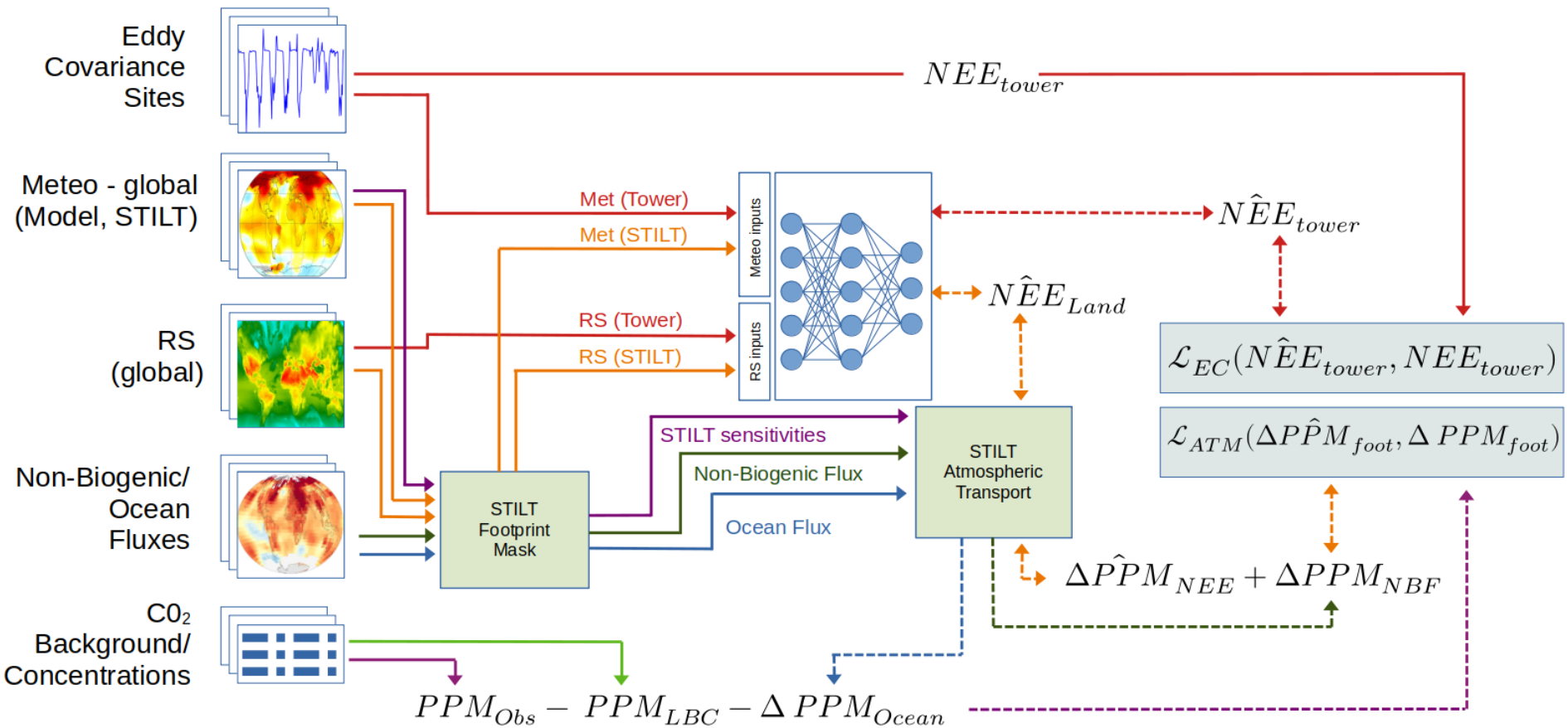
Xu et al., 2017, AFM

And now we can do ecology!



Allan Kaplan, UW-Madison

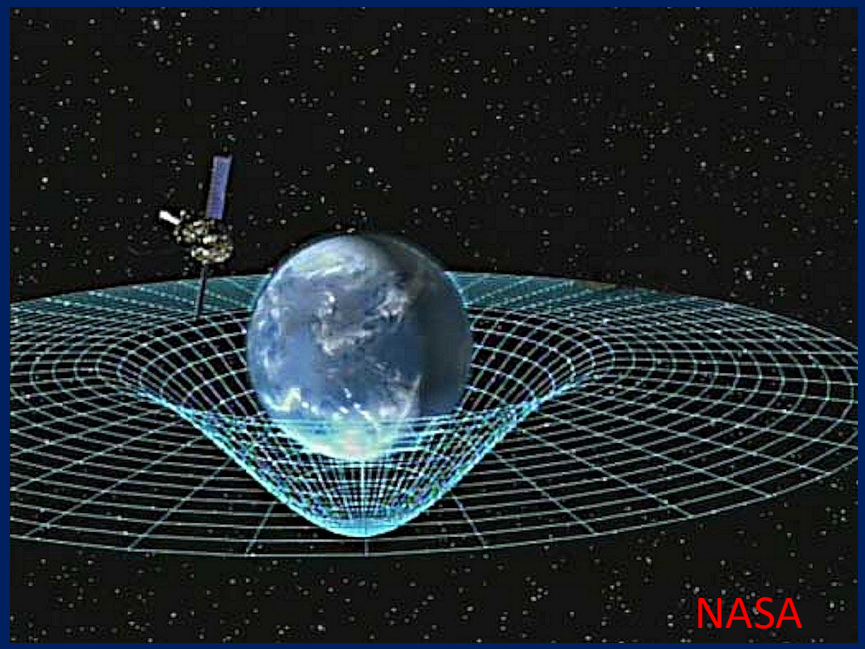
And build tools for convergence



Upton et al., 2026, ACP

Why now?

- We are entering an era where ecological observing technology and methods for transforming observations allow us to evaluate processes in multiple spatial and temporal scales at once



Caveat: Biology is not physics

A wise ecologist once told me,
"If Physics sets the laws, then
Biology is full of lawyers"

Some questions to ponder

- Can we develop community tools for properly down/up/right scale individual site tower fluxes in a world of hyper-everything remote sensing?
- Can we gain information from additional plot-level measurements and top-down observations (and rectify biases with towers)?
- How might this "information-gain" lead to better evaluation of Earth system models, quantification of Nature-based climate solutions, and decision-making around ecosystem resource management?

Go ponder the point!

Thanks
Ankur Desai, desai@aos.wisc.edu
Ecometeorology Lab
Collaborators
NSF, Google, USDA, DOE